

Graph bandwidth of weighted caterpillars[☆]

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Abstract

Graph bandwidth minimization (GBM) is a classical and challenging problem in graph algorithms and combinatorial optimization. Most of existing researches on this problem have focused on unweighted graphs. In this paper, we study the bandwidth minimization problem of weighted caterpillars, and propose several algorithms for solving various types of caterpillars and general graphs. More specifically, we show that the GBM problem on caterpillars with hair-length at most 2 and the GBM problem on star-shape caterpillars are NP-complete, and give a lower bound of the graph bandwidth for general weighted graphs. For caterpillars with hair-length at most 1, we present an $O(n \log n \log(nw_{\max}))$ -time algorithm to compute an optimal bandwidth layout, where n is the total number of vertices in the graph and $w_{\max}$ is the maximum wedge weight. For caterpillars with hair-length at most k , we give a k -approximation algorithm. For arbitrary caterpillars and general graphs, we give a heuristic algorithm and some experimental results. Experiments show that the solutions obtained by our heuristic algorithm are roughly within a factor of $c \log(n)$ of the lower bound for a small number c , which is consistent with the inapproximability results of this problem (i.e., no constant approximation for the GBM problem unless $P = NP$).

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1. Introduction

The graph bandwidth minimization (or GBM) problem for a given graph $G = (V, E)$ is to find a linear ordering (layout) of the vertices V such that the maximum difference between the labels (or rankings) of the two endpoints of any edge in E is minimized. The GBM problem is a fundamental problem in the fields of graph algorithms and combinatorial optimization, and finds applications in many other areas. It was originally used for modeling the problem of re-ordering the rows and columns of a sparse square matrix so that the non-zero entries of the matrix form a band along the diagonal with minimum width and consequently computations on such a matrix can be carried out more efficiently. Early study on this problem dates back to the 50s of last century, and since then extensive research has been done on this problem. Some good surveys on this problem can be found in papers [2,5].

The GBM problem in general is extremely difficult to solve optimally. It has been shown that the decision version of this problem is NP-complete for general graphs [13]. The problem remains to be NP-complete even for simple graphs such as trees with maximum degree 3 [7], caterpillars with hair-length no more than 3 [12], grid graphs and unit disk

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graphs [4]. Polynomial time solutions were found only for very special graphs such as caterpillars with hair-length no more than 2 [1], interval graphs [14], and chain graphs [11]. In 1998, Blache et al. showed that there is no PTAS for general graphs and trees [8] unless $P = NP$. Later, Unger further showed that it is NP-hard to approximate this problem within any constant factor [15].

Several approximation algorithms have been obtained for the GBM problem on general or special graphs. In [6], Feige presented an $O(\log^3 n \sqrt{\log n \log \log n})$ -approximation algorithm by using the powerful volume respecting embedding technique. Later, Gupta showed that the approximation ratio for trees can be improved to $O(\log^2 n \sqrt{\log n})$ by using similar techniques [9]. For caterpillars, Haralambides et al. presented a simple but elegant $O(\log n)$ -approximation algorithm [10].

So far most of the researches on the GBM problem have focused on unweighted graphs. Very few results were previously known for weighted graphs. Part of the reason is that the edge weights could dramatically complicate the optimization problem. The objective of the GBM problem on weighted graphs is to minimize the maximum weighted label difference of an edge (i.e., the edge weight times the label difference). In this paper, we study the GBM problem of weighted caterpillars.

Our problem is motivated by applications in VLSI layout, such as the legalization problem. In the legalization problem, a key task is to layout a graph on a rectangular area of a chip so that certain objective functions can be optimized. Each vertex in the graph is a smaller rectangle representing a modular of a circuit layout hierarchically, and the weight of an edge in the graph indicates the amount of connection (or wires) between the two corresponding modulars of the endpoints of the edge. The layout is often completed in a row-by-row fashion and the optimization task in each row can be modeled as a GBM problem on a weighted graph.

Compared to the unweighted counterpart, the GBM problem on weighted caterpillars is much harder to solve. Monien showed that the GBM problem for unweighted caterpillars with hair-length at most 3 is NP-complete [12], and Assmann et al. gave an algorithm to find the optimal layout for unweighted caterpillars with hair-length 1 and 2 [1]. For weighted caterpillars, we show that the GBM problem is NP-complete even for caterpillars with hair-length at most 2 and for star-shape caterpillars. (The star-shape caterpillars can also be solved in polynomial time when there is no weight on edges.)

In this paper, we have the following main results on weighted graph bandwidth minimization problem: we give a lower bound for general weighted graphs. For caterpillars with hair-length at most 1, we present an $O(n \log n \log(nw_{\max}))$ -time algorithm to compute an optimal bandwidth layout, where n is the total number of vertices in the graph and $w_{\max}$ is the maximum edge weight. The algorithm in [1] for unweighted caterpillars cannot guarantee an optimal solution for this case. For caterpillars with hair-length at most k , we make use of the lower bound and the algorithm for caterpillars with hair-length 1 to derive a k -approximation algorithm. For arbitrary caterpillars and general graphs, we give a heuristic algorithm. Experiments on a large set of randomly generated weighted graphs and caterpillars show that the solutions obtained by our heuristic algorithm are roughly within a factor of $c \log(n)$ of the lower bound for a small number c . This matches the so far best approximation ratio (i.e., $O(\log n)$) for unweighted caterpillars in [10] and is consistent with the inapproximability results of the GBM problem in general graphs (i.e., there is no constant approximation for the GBM problem unless $P = NP$).

The rest of the paper is organized as follows. In Section 2, we give some preliminaries of the problem. In Section 3, we give a lower bound for weighted general graphs. Section 4 describes our algorithm for caterpillars with hairlength at most 1. Hardness results of the weighted caterpillars are presented in Sections 5 and 6. The k -approximation algorithm for caterpillars with hair-length at most k is given in Section 7. Heuristic algorithm and experimental results for weighted general graphs and caterpillars are discussed in Section 8.

2. Preliminaries

Let $G = (V, E)$ be a weighted graph with weights $w : E \rightarrow \mathbb{Q}^+$ and $|V| = n$. A *linear layout* or simply a *layout* L of G is an ordering of V with a bijective labeling function $L : V \rightarrow [n] = \{1, 2, \dots, n\}$. We say that G has bandwidth B under the layout L , denoted by $b_L(G) \leq B$, if $|L(u) - L(v)| \times w(u, v) \leq B$ for every edge $(u, v) \in E$. The bandwidth $b(G)$ of G is the minimum bandwidth under all possible layouts, i.e., $b(G) = \min_L \{b_L(G) | L \text{ is a layout of } G\}$. We assume that G is connected, since otherwise the bandwidth of G is simply that of the connected component with the largest bandwidth.

Let $e \in E$ be an edge and $w(e)$ be the weight of e . We define *distance* of e as $d(e) = 1/w(e)$. The length $d(P)$ of a path P is the sum of all distances of edges on this path. The *diameter* $d(G)$ of G is defined as the distance of the longest shortest path between any pair of vertices in G .

A *caterpillar* C is a tree which has a simple path, called *backbone*, and various other appendage line graphs attached to the vertices of the backbone. Each line graph is called a *hair*. A hair of the caterpillar is called an h_v subtree if it is attached to a backbone vertex v . The hair-length of h_v is the number of edges of h_v and the hair-length of C is the maximum hair-length of all its hairs. Let u be a vertex in an h_v subtree and x be the length (i.e., the number of edges) of path $v \rightsquigarrow u$. Let u' be another vertex in the same subtree h_v . We say u' is the parent of u if the length of the path $v \rightsquigarrow u'$ is $x - 1$.

3. Lower bound of general weighted graphs

In the section, we give a lower bound for the GBM problem on general weighted graphs.

Lemma 1. *For any connected subgraph G' of G , there exists a pair of vertices v_1, v_t , such that for any path $P = v_1 v_2 \dots v_t$ in G' ,*

$$\sum_{i=1}^{t-1} \left\lfloor \frac{b(G')}{w(v_i v_{i+1})} \right\rfloor \geq |G'| - 1,$$

where $|G'|$ is the number of vertices in G' .

Proof. Let L be an optimal layout of G' with bandwidth $b(G')$. Let v_1 be the first vertex in L and v_t be the last vertex in the layout L . Consider any path $P = v_1 v_2 \dots v_t$ in G' that connects v_1 and v_t . Let the labels of the vertices in P be $L(v_1), L(v_2), \dots, L(v_t)$, respectively. Since v_1 and v_t are the first and last vertices in the linear layout of G' , we have $\sum_{i=1}^{t-1} |L(v_{i+1}) - L(v_i)| \geq |G'| - 1$. For each edge (v_i, v_{i+1}) in P , its bandwidth $|L(v_{i+1}) - L(v_i)| \times w(v_i v_{i+1}) \leq b(G')$, which implies $|L(v_{i+1}) - L(v_i)| \leq \lfloor b(G')/w(v_i v_{i+1}) \rfloor$ since all labels $L(v_1), L(v_2), \dots, L(v_t)$ are integer numbers. Thus, we have

$$\sum_{i=1}^{t-1} \left\lfloor \frac{b(G')}{w(v_i v_{i+1})} \right\rfloor \geq \sum_{i=1}^{t-1} |L(v_{i+1}) - L(v_i)| \geq |G'| - 1. \quad \square$$

Lemma 2. *Let G' be any connected subgraph of G . The bandwidth $b(G) \geq \max_{G'} |G'| - 1/d(G')$.*

Proof. For a subgraph G' , let v_1 and v_t be the pair of vertices satisfying Lemma 1. Let $P = v_1 v_2 \dots v_t$ be the shortest path between v_1 and v_t in G' . By Lemma 1,

$$\sum_{i=1}^{t-1} \left\lfloor \frac{b(G')}{w(v_i v_{i+1})} \right\rfloor \geq |G'| - 1.$$

Hence,

$$\sum_{i=1}^{t-1} \frac{b(G')}{w(v_i v_{i+1})} \geq |G'| - 1.$$

Thus,

$$b(G') \geq \frac{|G'| - 1}{\sum_{i=1}^{t-1} \frac{1}{w(v_i v_{i+1})}} = \frac{|G'| - 1}{\sum_{i=1}^{t-1} d(v_i v_{i+1})}.$$

Since $\sum_{i=1}^{t-1} d(v_i v_{i+1}) \leq d(G')$, we have

$$b(G') \geq \frac{|G'| - 1}{d(G')}.$$

The lemma follows from the fact that the bandwidth of G is no less than that of its any subgraph G' . $\square$

4. Algorithm for caterpillars with hair-length at most 1

Let C_1 be a caterpillar with hair-length no more than 1. We consider the decision version of this problem. That is, given a positive integer b , decide whether there exists a layout L such that $b_L(C_1) \leq b$. Note that although the bandwidth is a rational number, we can always scale up the weights and make it an integer. Below we first present an algorithm for this decision problem. Our algorithm takes as input a caterpillar C_1 with hair-length no more than 1 and an integer b and tries to find a layout with bandwidth no more than b . We then show that if the algorithm fails to find such a layout, then there is a subgraph C'_1 of C_1 that violates Lemma 1. Thus, $b(C_1) \geq b(C'_1) > b$.

We say a number in $[1, |C_1|]$ is *free* if no vertex is permanently labeled with it. Let $L(v)$ be the label of a vertex v . Let u be a hair vertex attached to a backbone vertex v . A possible label for u should be a free number in $[L(v) - \lfloor b \times d(u, v) \rfloor, L(v) + \lfloor b \times d(u, v) \rfloor]$, where $d(u, v)$ is the distance between u and v . Let $D = v_1 v_2 \dots v_r$ be the backbone of C_1 . Our algorithm has the following main steps.

Algorithm 1.

- (1) Label permanently the vertices along the backbone D with numbers $0, \lfloor b \times d(v_1, v_2) \rfloor, \lfloor b \times d(v_1, v_2) \rfloor + \lfloor b \times d(v_2, v_3) \rfloor, \dots, \sum_{i=1}^{r-1} \lfloor b \times d(v_i, v_{i+1}) \rfloor$.
- (2) For each hair vertex u attached to some backbone vertex v , label it temporarily with the number $L(v) + \lfloor b \times d(u, v) \rfloor$. Note that multiple vertices may be labeled (permanently or temporarily) with the same number.
- (3) Sort vertices in the non-decreasing order of their labels, and break ties arbitrarily.
- (4) Scan all hair vertices in the above order and label permanently each hair vertex u with the smallest free number in the range $[L(v) - \lfloor b \times d(u, v) \rfloor, L(v) + \lfloor b \times d(u, v) \rfloor]$, where v is the parent of u . If there is no free number for u , then stop and return false.

Below we show that the above algorithm solves the decision problem of the GBM problem on caterpillars with hair-length at most 1.

Theorem 3. *If the above algorithm fails to find a layout L of C_1 with $b_L(C_1) = b$, then $b(C_1) > b$.*

Proof. Suppose the algorithm returns false. We will show that there exists a subgraph C'_1 of C_1 that violates Lemma 1.

In the above algorithm, a failure occurs only in Step 4 when it tries to label some hair vertex, say u , with a temporary label l' . Obviously, there must exist at least another vertex with label l' , otherwise u can be labeled permanently with l' .

When the failure occurs, let $L' = \{l_1, l_2, \dots, l_t = l'\}$ be the longest consecutive sequence of numbers that are occupied (i.e., each $l_i \in L'$ has been assigned as a permanent label to some vertex u_i and $l_1 - 1$ is free). Let $V' = \{u_1, u_2, \dots, u_t, u\}$. We show that the induced subgraph $C'_1 = (V', E')$ violates Lemma 1.

We note that the following properties are true for vertices in V' .

- (1) Each $u_i, i \in [1, t]$ is either a backbone vertex or a hair vertex attached to some backbone vertex.
- (2) If $u_i, i \in [1, t]$ is a hair vertex and v_i is its parent, then $v_i \in V'$.

To show that the second property is true, we need to prove that $l_1 \leq L(v_i) \leq l'$. To see $L(v_i) \leq l'$, we note that we always assign to hair vertices temporary labels no less than the permanent labels of their parents and scan hair vertices based on the non-decreasing order of their temporary labels. The temporary label of u_i must be no larger than that of u (i.e., l'), and hence $L(v_i) \leq l'$. To show $l_1 \leq L(v_i)$, we assume that $l_1 > L(v_i)$. Then in the range $[L(v_i) - \lfloor b \times d(u_i, v_i) \rfloor, L(v_i) + \lfloor b \times d(u_i, v_i) \rfloor]$, we have a free number $l_1 - 1 < L(u_i)$ which can be used to label u_i . This contradicts the fact that we always find the smallest free label for each hair vertex in its range.

The above two properties implies that C'_1 is a connected graph since by Step 1, backbone vertices in C'_1 are always connected.

Let $v_h, v_{h+1}, \dots, v_k, \dots, v_g$ be backbone vertices in V' . Suppose u_1 is connected to v_x and u is attached to v_y . Let $P = u_1 v_x v_{x+1} \dots v_z u_t$ be a path (see Fig. 1). Let $d'(v_i, v_j) = \sum_{l=i}^{j-1} \lfloor b \times d(v_l, v_{l+1}) \rfloor$. We have the following claim.

Claim. $\lfloor b \times d(u_1, v_x) \rfloor + d'(v_x, v_z) + \lfloor b \times d(v_z, u_t) \rfloor$ is the maximum over the distances of all paths in C' .

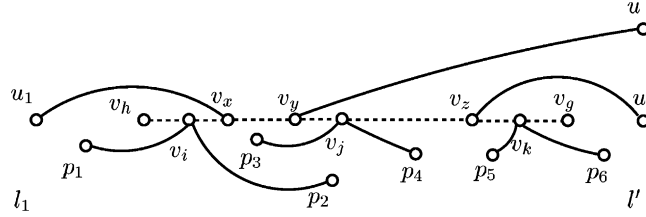


Fig. 1. An ordering of vertices of a subgraph C'_1 (vertices are ordered from left to right).

To prove the above claim, we first consider an arbitrary hair h whose both endpoints are in V' . h can be attached to either a backbone vertex v_i between v_h and v_x , a backbone vertex v_j between v_x and v_z , or a backbone vertex v_k between v_z and v_g (see Fig. 1). For each of the three cases, we have two subcases depending on whether the hair vertex of h has a smaller or greater label than its parent (i.e., the backbone vertex). Thus, we have in total six types of hairs, p_1v_i , p_2v_i , p_3v_j , p_4v_j , p_5v_k and p_6v_k .

For hair p_1v_i , we have $\lfloor b \times d(p_1, v_i) \rfloor + d'(v_i, v_x) \leq \lfloor b \times d(u_1, v_x) \rfloor$, otherwise p_1 would be layouted to the left of u_1 by Step 4 of the algorithm. Also $\lfloor b \times d(p_1, v_i) \rfloor \leq d'(v_i, v_z) + \lfloor b \times d(v_z, u_t) \rfloor$, otherwise p_1 will not be relabeled before u_t .

Similarly, we also have the following inequality for hairs p_3v_j and p_5v_k :

$$\begin{aligned} \lfloor b \times d(p_3, v_j) \rfloor &\leq \lfloor b \times d(u_1, v_x) \rfloor + d'(v_x, v_j), \\ \lfloor b \times d(p_3, v_j) \rfloor &\leq d'(v_j, v_z) + \lfloor b \times d(v_z, u_t) \rfloor, \\ \lfloor b \times d(p_5, v_k) \rfloor &\leq \lfloor b \times d(u_1, v_x) \rfloor + d'(v_x, v_k), \\ \lfloor b \times d(p_5, v_k) \rfloor + d'(v_z, v_k) &\leq \lfloor b \times d(v_z, u_t) \rfloor. \end{aligned}$$

Similar results hold for hairs p_2v_i , p_4v_j and p_6v_k .

Putting all the inequalities together, we can show that $\lfloor b \times d(u_1, v_x) \rfloor + d'(v_x, v_z) + \lfloor b \times d(v_z, u_t) \rfloor$ is the maximum over all paths in C'_1 by considering all types of paths. Below we only consider one type of paths $P' = p_1v_i v_{i+1} \dots v_k p_5$. The other types of paths can be proved similarly.

$$\begin{aligned} &\lfloor b \times d(p_1, v_i) \rfloor + d'(v_i, v_k) + \lfloor b \times d(p_5, v_k) \rfloor \\ &= \lfloor b \times d(p_1, v_i) \rfloor + d'(v_i, v_x) + d'(v_x, v_z) + d'(v_z, v_k) + \lfloor b \times d(p_5, v_k) \rfloor \\ &\leq \lfloor b \times d(u_1, v_x) \rfloor + d'(v_x, v_z) + \lfloor b \times d(v_z, u_t) \rfloor. \end{aligned}$$

Since all numbers in $[l_1, l_t]$ are occupied, we have $|C'_1| = l' - l_1 + 2$

$$\lfloor b \times d(u_1, v_x) \rfloor + d'(v_x, v_z) + \lfloor b \times d(v_z, u_t) \rfloor = l' - l_1 = |C'_1| - 2.$$

Thus there is no path that satisfies Lemma 1, which implies that the graph bandwidth of C'_1 is larger than b . This concludes the proof of the theorem. $\square$

Corollary 4. *There is an $O(n \log n \log(nw_{\max}))$ -time algorithm for finding the bandwidth of C_1 , where $w_{\max}$ is the maximum edge weight of C_1 .*

Proof. Clearly, the bandwidth of C_1 is between $w_{\max}$ and $n \times w_{\max}$. We can use the above algorithm for the decision problem to do a binary search on the interval $[w_{\max}, n \times w_{\max}]$ to find $b(C_1)$.

Note that, Steps 1 and 2 take $O(n)$ time and Step 3 takes $O(n \log n)$ time. We can use a balanced binary tree to maintain the free numbers. Thus labeling a vertex only takes $O(\log n)$ time. Therefore, the whole algorithm for the decision problem takes $O(n \log n)$ time. Thus, the total time to find $b(C_1)$ is $O(n \log n \log(nw_{\max}))$. Once we have a labeling of C_1 , it can be easily changed to a layout by ordering the vertices according to the order of their labels. $\square$

Remarks. Even though the running time of our algorithm depends on the input $w_{\max}$, the algorithm is still a fully polynomial time algorithm since it only logarithmically depends on $w_{\max}$ (i.e., it linearly depends on the number of bits of the input). Also it is possible to further improve the running time by using the parametric search technique.

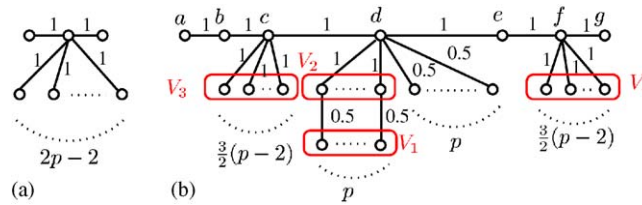


Fig. 2. (a) A barrier of height p . (b) A turning point of height p , $p \equiv 0 \pmod{2}$.

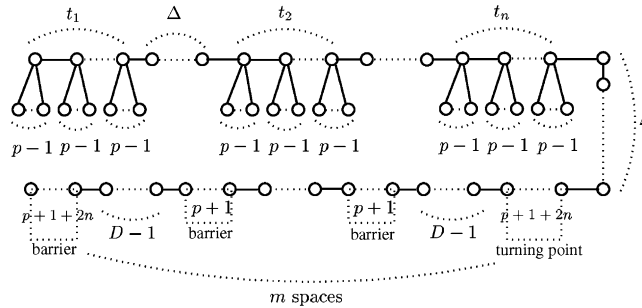


Fig. 3. The caterpillar C_2 , $\Delta = 2\{m(D+2) - 2\}$, p has to be chosen in an appropriate way.

5. Caterpillars with hair-length at most 2

In this section, we show that the GBM problem on caterpillars with hair-length at most 2 is NP-complete.

Theorem 5. *The GBM problem on caterpillars with hairs-length at most 2 is NP-complete.*

Clearly, the problem is in NP since for a graph $G = (V, E)$, a weighting function $w : E \rightarrow \mathbb{Q}^+$ and a value b , for each proof (i.e., a layout of V) we can calculate the bandwidth of this layout in $O(|E|)$ time and then verify whether it is no more than b . To prove the NP-hardness of the problem, we follow the approach used in [12] for proving the NP-hardness for unweighted caterpillars and reduce the Multiprocessor Scheduling problem to our GBM problem. Let C_2 be a weighted caterpillar with hair-length at most 2. Given a set $T = \{t_1, t_2, \dots, t_n\}$ of tasks (t_i is the execution time of the i th task), a deadline D , and m processors, we construct a caterpillar C_2 (see Fig. 3) and an integer b such that C_2 has bandwidth b if and only if the tasks in T can be scheduled on the m processors no later than the deadline D . The Multiprocessor Scheduling problem is strongly NP-complete and therefore we can assume that all the t_i are polynomially bounded in n .

The main idea of the construction is to simulate the scheduling. We use some portions of C_2 to simulate tasks and some portions to simulate processors, called *task portions* and *processor portions*, respectively. More specifically, each task t_i is represented by a caterpillar C_1^i with t_i backbone vertices and $p-1$ hairs (of length 1) on each backbone vertex for some parameter p , and each processor is represented by a caterpillar with $D-1$ backbone vertices and no hair (see Fig. 3). We also use two special caterpillars, called *barrier* and *turning point* (see Fig. 2), to separate the task portions and processor portions. A barrier is used to separate two processors portions such that a task portion cannot be assigned to multiple processor portions. The task portions and processor portions are separated by a turning point. The turning point ensures that in every optimal layout, the task portions and processor portions are always on the same side of the turning point so that vertices representing tasks can be layouted between the backbone vertices of the processors (Fig. 3).

As shown in [12], in order to make the reduction work, a key difficulty is to ensure that the barrier and turning point have the following properties. (1) Each barrier of height p and each turning point of height p (see Fig. 2) have bandwidth p ; (2) in every optimal layout of the turning point, the first and last backbone vertices (i.e., a and g in Fig. 2) are assigned to the same half of the layout.

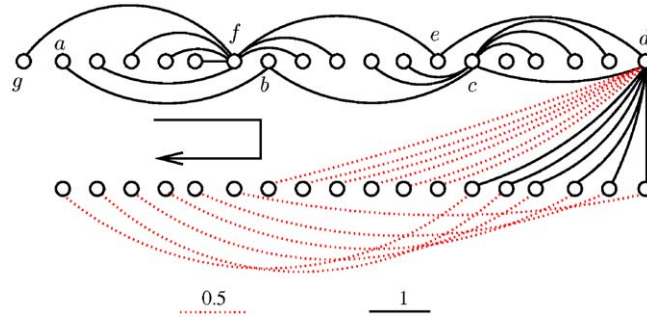


Fig. 4. A layout of a turning point.

To ensure the above properties, we modify the structure of the turning point in [12] by assigning different weights to its edges and shortening its hair-length from 3 to 2. (Note that the other part of the constructed caterpillar has hair-length 1.) Let $T_p = (V, E)$ denote the turning point of height p . The following lemmas show that the turning point has the expected properties.

Lemma 6. *The bandwidth of the turning point T_p in Fig. 2(b) is p .*

Proof. T_p has exactly $6p + 1$ vertices. The diameter of T_p is clearly 6. By Lemma 2, we know that the bandwidth of T_p is no less than $(6p + 1 - 1)/6 = p$. Fig. 4 gives a layout of T_p with bandwidth exactly p . Thus, the lemma follows. $\square$

Lemma 7. *Let L be an optimal layout of the turning point $T_p = (V, E)$ with bijective function $L : V \rightarrow \{1, \dots, 6p + 1\}$ (i.e., $|L(i) - L(j)|w(i, j) \leq p$ for all $(i, j) \in E$) and $p \geq 6$. Then either $L(a), L(g) < 3p + 1$ or $L(a), L(g) > 3p + 1$.*

Proof. Consider the path $P = p_1 \dots p_k$ that connects the leftmost vertex p_1 and the rightmost vertex p_k in L . The length (in terms of number of edges) of P can be 5 or 6 only. This is because in T_p , a path can have either 0, 1, or 2 edges with weight 0.5. By a similar argument given the proof of Lemma 1 we know that for the path with 0 or 1 edge of weight 0.5, the number of edges for a path to span $6p + 1$ vertices and with bandwidth p has to be at least 5 or 6. From the structure of T_p , it is easy to see that the length of a path with 2 0.5-weight edges is 3. By the same argument given in the proof of Lemma 1, we know that the bandwidth will be at least $6p/5$, contradicting the fact that L is an optimal layout.

The only path with length 6 in T_p is $a - b - c - d - e - f - p_k$, $p_k \in \{g\} \cup V_4$ and the weight of each edge on this path is 1. Since $|L(p_k) - L(p_1)| = 6p$ and $|L(u) - L(v)| \leq p$, $\forall (u, v) \in P$, we have $L(d) = 3p + 1$. If the length of P is 5, one of p_1 and p_k must be in V_1 . Suppose $p_1 \in V_1$. Since all the edges on P except $p_1 p_2$ have weight 1 and $w(p_1 p_2) = 0.5$, we also have $L(d) = 3p + 1$.

Suppose $L(a) < 3p + 1$ and $L(g) > 3p + 1$, we show below that there will be a contradiction. Since $L(d) = 3p + 1$, the vertices that can have labels $1, \dots, p$ must be a or in V_1 . We will show that $L(a) \in [1, p]$. If not, then $[1, p]$ are all occupied by V_1 and $[2p + 1, 3p]$ are all occupied by V_2 . Then $L(a), L(b) \in [p + 1, 2p]$, but c cannot be labeled. So $[1, p]$ must be occupied by a and $p - 1$ vertices from V_1 (Fig. 5). Then we have $L(b) \in [p + 1, 2p]$ and $L(c) \in [2p + 1, 3p]$. The other $p - 1$ labels in $[2p + 1, 3p]$ must be occupied by $p - 1$ vertices from V_2 . Then the vertices in V_3 can have labels within $[L(c) - p, L(c) + p]$. The number of free numbers in $[L(c) - p, L(c) + p]$ is $2p + 1 - 3 - (p - 1) = p - 1$. But we have $\frac{3}{2}(p - 2)$ vertices in V_3 which implies $\frac{3}{2}(p - 2) \leq p - 1$. We get $p \leq 4$ contradicts to the assumption that $p \geq 6$. $\square$

With the above lemmas, we can construct the caterpillar C_2 from any instance $\mathcal{Y} = (\{t_1, \dots, t_n\}, D, m)$ of the Multiprocessor Scheduling problem as in Fig. 3 and apply the following two lemmas (proved in [12]) to complete the reduction. This concludes the proof of Theorem 5.

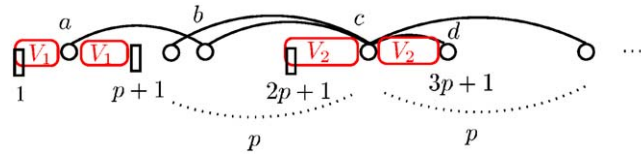


Fig. 5. An optimal layout of the turning point.

Lemma 8. If $\mathcal{Y}$ has a solution then C_2 has bandwidth $p + 1 + 2n$.

Proof. See [12]. $\square$

Lemma 9. If $p > 2n(D + 4)$ and if C_2 has bandwidth $p + 1 + 2n$, then $\mathcal{Y}$ has a solution.

Proof. See [12]. $\square$

6. Star-shape caterpillars

In this section, we first show that for unweighted star-shape caterpillars, the GBM problem can be solved optimally in polynomial time. Then we show that the GBM on general star-shape caterpillars is NP-complete. Let us start with unweighted star-shape caterpillars (Fig. 6).

Algorithm 2.

- (1) Label the center (root) v of the caterpillar with number 0.
- (2) Label half of v 's children with consecutive positive integer numbers starting from 1; label the other half with consecutive negative integer numbers starting from -1 .
- (3) Label the nodes of depth i ($i = 2, 3, \dots$) with consecutive integer numbers before labeling any nodes of depth $i + 1$, and the labels of the nodes of depth i follow the same order as their parents.
- (4) Map all the labels to their corresponding integer numbers in $[1, |V|]$.

Theorem 10. For an unweighted star-shape caterpillar $G = (V, E)$, the GBM problem can be solved in $O(|V|)$.

Proof. Consider a subgraph that contains the root v and all of its children S . By Lemma 2, $b(G) \geq |S|/2$. Since $b(G)$ is an integer number, we have $b(G) \geq \lceil |S|/2 \rceil$. It is easy to see that the bandwidth of the layout produced by Algorithm 2 equals to $\lceil |S|/2 \rceil$ and Algorithm 2 runs in time $O(|V|)$. $\square$

Theorem 11. The GBM problem on star-shape caterpillars is NP-complete.

Proof. Clearly, this problem is in NP. Similar to the NP-completeness proof of Theorem 5, we reduce from the Multiprocessor Scheduling problem. Given a set $S = \{t_1, t_2, \dots, t_n\}$ of tasks (t_i is the execution time of i th task), a deadline D and m processors, we shall construct a star-shape caterpillar C_s and an integer b such that C_s has bandwidth b if and only if the tasks in T can be scheduled on the m processors satisfying the deadline D . We can assume that all the t_i s are polynomially bounded in n .

We construct C_s as follows. We first build the backbone. Let $T = \sum_{i=1}^n t_i$. We put a set B_1 of $T + 1$ nodes to the left. Then we connect a set M of m nodes and a set B_2 of $T + m + 1$ nodes to the right. Corresponding to each task i , we construct a hair with t_i nodes. We associate edges with weights as in Fig. 7. Each edge with its left endpoint in M represents a processor.

Let $b = \beta$, $\alpha = \beta/(T + m + 1)$. We first show that if there is a feasible scheduling for the Multiprocessor Scheduling problem, then we have a layout of C_s with bandwidth β . We construct the layout by assigning all the nodes of each task t_i (or hair) to the edge corresponding to the processor executing t_i in the scheduling (i.e., layout all the nodes of t_i between the two endpoints of the processor). Clearly, the bandwidth of this layout is β . Then we show that if there is a layout with bandwidth β , then there is a feasible scheduling. Note that no hair node can be put outside of the backbone

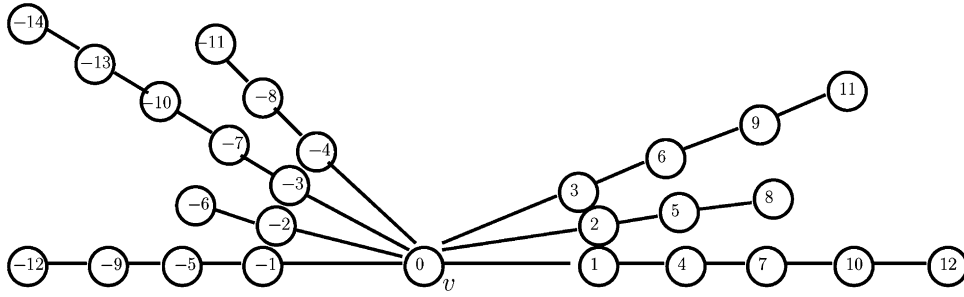
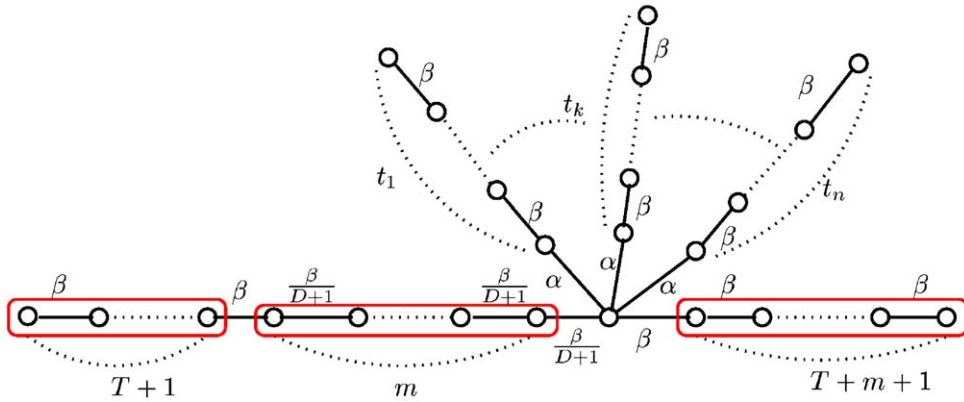


Fig. 6. The labeling of an unweighted star-shape caterpillar after Step 3.

Fig. 7. The caterpillar C_s .

and all nodes of a hair cannot be split (otherwise one of the edge on this hair will have a bandwidth larger than β). Thus, all nodes of each hair can be put on one and only one edge of a processor (in M). The corresponding scheduling of this layout will meet the deadline D and is therefore feasible. $\square$

7. Approximation algorithm for caterpillars with hair-length at most k

Let C_k be a weighted caterpillar with hair-length at most k . The following simple algorithm yields a layout for C_k .

Algorithm 3.

- (1) Let C'_k be the maximal subgraph of C_k such that C'_k is a caterpillar with hair-length at most 1.
 - (2) Apply Algorithm 1 to C'_k and find an optimal layout L' .
 - (3) Put all the descendants of each hair vertex u of C'_k to its left neighboring positions in L' while keeping their orders.
- Return the resulting layout L .

Theorem 12. Algorithm 3 is a k -approximation algorithm for the GBM problem on weighted caterpillars with hair-length at most k .

Proof. Let $OPT' = b_{L'}(C'_k)$ and OPT be the bandwidth of C_k . Clearly $OPT \geq \max(OPT', w_{\max})$. By the algorithm,

$$b_L(C_k) \leq \max(kOPT', w_{\max}) \leq kOPT. \quad \square$$

8. Heuristic algorithm for arbitrary caterpillars and general graphs

In this section, we present a heuristic algorithm for arbitrary caterpillars and general graphs and show some experimental results. Our algorithm is a generalization of the CutHill–McKee algorithm [3].

The CutHill–McKee algorithm is for unweighted graphs. The algorithm starts from a vertex with small degree, and put its neighbors to level one. Repeat the procedure for vertices in each level such that all adjacent vertices have been put in neighboring or the same level. It has been shown that the bandwidth is no more than twice of the size of largest level.

We extend the idea of the CutHill–McKee algorithm to weighted graphs. In weighted graphs, we have to consider the weights of edges. If we always put unvisited adjacent vertices of each vertex into its neighboring levels, then the size of some level will become unnecessarily large. To fix this problem, we compute the level of a vertex as follows. Let $e = (u, v)$ be an edge with weight $w(e)$ and $lvl(u)$ be the level of u . We compute the level $lvl(v)$ of v as $lvl(v) = \min\{lvl(v), lvl(u) + w_{\max}/w(e)\}$. Let $d_{\max}$ (or $d_{\min}$) be the maximum (or minimum) degree of the graph. Let $d = (d_{\max} + d_{\min})/2$. Our algorithm runs as follows.

Algorithm 4.

- (1) For each vertex v , set $lvl(v) = \infty$.
- (2) For each vertex u with degree $\leq d$ do
 - (a) Select u as the starting vertex. Set $lvl(u) = 1$.
 - (b) For each neighbor v of u , set $lvl(v) = \min(lvl(v), lvl(u) + w_{\max}/w(u, v))$.
 - (c) The procedure is repeated for each node at levels in the increasing order.
 - (d) Label vertices level by level. Store the bandwidth of the layout.
- (3) Output the minimum bandwidth of all computed layouts.

The above algorithm runs in $O(n^3)$, where n is the number of vertices in the graph.

To examine the performance of our algorithm, we implement it in C++ and test it with a large set of randomly generated graphs and caterpillars whose number n of vertices are $25i$ for $1 \leq i \leq 80$. For each $n = 25i$, we generated 10 different general graphs and 10 different caterpillars. Our experiments are conducted on a Dell PC with an Intel Pentium 4 2.26 GHz CPU and running Linux 8.0. Since it is impossible to determine the optimal solution for each randomly generated graph, we compare our results with the lower bound given in Lemma 2. The ratio (i.e., the computed bandwidth over the lower bound) for each different n is the average ratio over the 10 different graphs (of the same n).

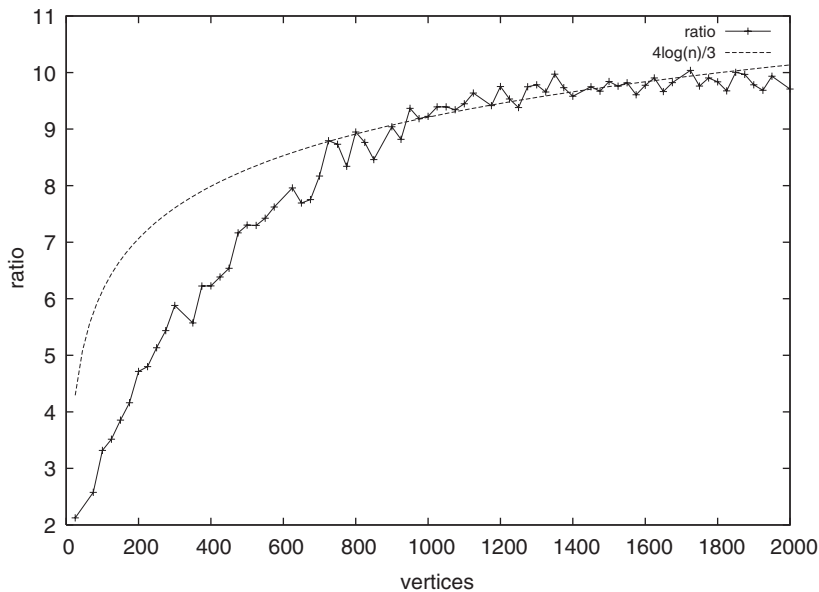


Fig. 8. Comparison between our generated bandwidth and the lower bound on random general graphs.

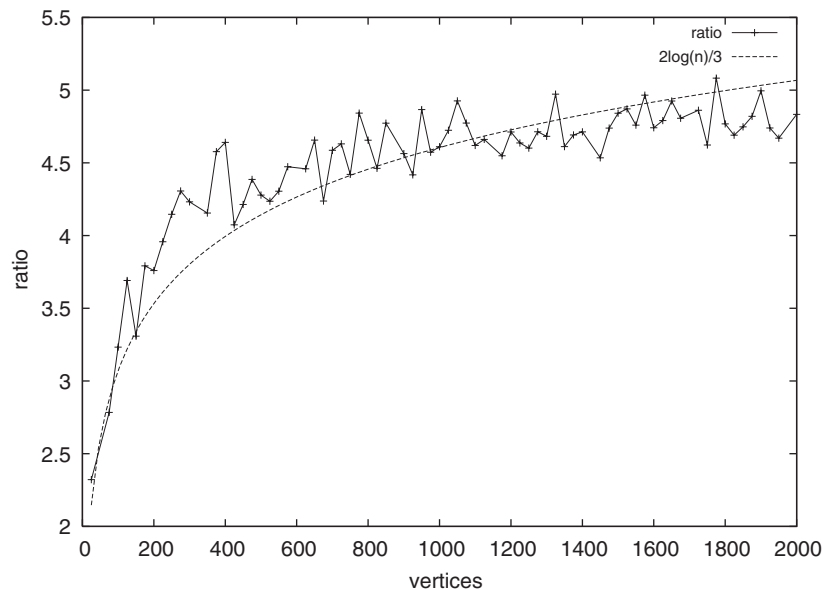


Fig. 9. Comparison between our generated bandwidth and the lower bound on random caterpillars.

Fig. 8 shows the comparisons of general graphs. The figure seems to suggest that the ratio of our computed bandwidth over the lower bound is roughly a $\frac{4}{3} \log(n)$ function of the number of vertices. When the number of vertices is less than 500, the ratio is smaller than $\frac{4}{3} \log(n)$. Fig. 9 shows the comparisons of caterpillars. The figure seems to suggest that the ratio of our computed bandwidth over the lower bound is roughly a $\frac{2}{3} \log(n)$ function of the number of vertices. We also notice that the computed bandwidth is smaller on caterpillars than on general graphs with the same number of vertices. We attribute this difference to the fact that caterpillars are trees and have much less edges to constrain the vertices. Experiments on the two types of randomly generated graphs match the so far best approximation ratio (i.e., $O(\log n)$) for unweighted caterpillars in [10] and is consistent with the inapproximability results of the GBM problem in general graphs (i.e., there is no constant approximation for the GBM problem unless $P = NP$).

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